HARSH AGARWAL

Computer Science Student

+91 8003479287 | ha2966311@gmail.com | linkedin.com/in/harsh-agarwal-891945318/ | github.com/harsh371-coder

SUMMARY

Computer Science undergraduate (B.Tech, 2028) at Bennett University. Proficient in Python, Java, SQL, and Machine Learning libraries including TensorFlow, Scikit-learn, and Pandas, AI Engineer. Experienced in building deep learning projects, Gen AI projects and full-stack Java applications, with hands-on expertise in collaborative development using Git and GitHub.

EDUCATION

Bachelor of Technology in Computer Science and Engineering *Aug 2024 – Aug 2028*

Bennett University, Greater Noida **CGPA: 8.65 / 10.0**

SKILLS SUMMARY

- **Languages:** Python, Java, SQL
- **Frameworks / Libraries:** TensorFlow, Keras, Scikit-learn, Pandas, NumPy, Matplotlib
- **Tools:** Git, GitHub, Power BI, Excel, MySQL, SQLite
- **Platforms:** PyCharm, Jupyter Notebook, Visual Studio Code, Google Colab
- **CS Fundamentals:** Data Structures & Algorithms, OOP, DBMS, Machine Learning, Deep Learning, Generative AI

PROJECTS

MediScan AI — Deep Learning MRI Segmentation System *2025 – 2026*

Tech Stack: *Python, TensorFlow, Keras, DeepLabV3+, U-Net, NumPy, Jupyter Notebook* | *Bennett University Research Project*

- Built an end-to-end deep learning system for automated MRI tumor segmentation across 3 organ datasets (brain, cardiac, prostate) using Python, TensorFlow, Keras, and NumPy on NIfTI/DICOM medical image formats.
- Trained and benchmarked 5 CNN architectures — U-Net, ResNet-U-Net, SegNet, DeepLabV3+ Baseline, and DeepLabV3+ with Transformer-Attention — achieving a best Dice Score of 0.87 and IoU of 0.79 on brain tumor segmentation.
- Evaluated all models across 6 standardized metrics (Dice Coefficient, IoU, Precision, Recall, F1-Score, Accuracy) on an 80/10/10 train-validation-test split, with data augmentation (rotation, flipping, zoom) applied to reduce overfitting.
- Developed a browser-based inference interface enabling clinicians to upload MRI scans and receive real-time segmentation masks, automated risk classification (Low/Medium/High), and downloadable diagnostic reports in under 2 seconds per scan.

Multi-Agent AI Research System — Automated Research & Report Generation Pipeline *2025 – 2026*

Tech Stack: *Python, LangChain, LangGraph, Mistral AI, Tavily API, Streamlit* | *Live Demo*

- Architected an end-to-end multi-agent research pipeline in Python using LangChain and LangGraph, orchestrating four autonomous agents — Search, Reader, Writer, and Critic — to turn a single topic into a structured, fact-checked report.
- Built a ReAct-style Search Agent and Reader Agent with LangGraph's tool-calling framework, integrating the Tavily Search API to retrieve and scrape the most relevant, up-to-date web sources for any research topic.
- Developed Writer and Critic chains powered by Mistral AI to draft structured research reports and automatically critique them for gaps and accuracy before final delivery.
- Designed and deployed a real-time Streamlit interface with live pipeline status tracking and downloadable reports, publicly hosted for end-to-end use by any visitor.

CERTIFICATIONS

- **Core Java Programming Revisited** – Infosys *(Mar 2025)*
- **Explore Machine Learning Using Python** – Infosys *(Nov 2025)*